Hacettepe University
Department of Industrial Engineering

EMÜ414- Mathematical Programming and Computer Applications

2025-2026 Spring

Term Project – Second Deliverable

Deadline: May 19, 2026, 23:59

1. Team Formation

The project must be completed in the same teams formed for the first deliverable.

2. Scope of the Second Deliverable

The second deliverable is the final version of the term project. Each team is expected to complete the mathematical modeling, implement the model in Gurobi, conduct computational analysis, and present the final results in a written report and a short presentation.

3. Final Report Requirements

The final report must be written in English and prepared in the format of a short academic paper.

Depending on the nature of the project, the paper should include the following main components:

- Title, team information, and abstract
- Introduction and problem description
- Mathematical formulation, including the decisions, objective, constraints, assumptions, problem classification, and the general structure of the mathematical model
- Model notation, including parameters and decision variables
- Implementation and computational analysis, including:
  - computational results and validation
  - sensitivity analysis, if possible
- Discussion, conclusion, and references
- Appendix, including a team photo

The report should clearly present both the mathematical model and the computational findings. Figures and tables should be clearly labeled and referred to in the text.

The entire mini paper, including references and appendices, must not exceed 10 pages.

The report must be prepared in LaTeX.

You are strongly encouraged to use Overleaf, but any other LaTeX platform is also acceptable.

4. Presentation

Each team must also prepare a short presentation summarizing the project.

The presentation should briefly cover:

- the problem and motivation
- the mathematical model
- the computational approach
- the main results

Each team will have 5 minutes for the presentation.

5. Submission

Submit the second deliverable electronically through www.hadi.hacettepe.edu.tr .

Do not submit a printed copy.

Your final submission must include the following files:

- minipaper.pdf
- slides.pdf
- model.py
- input data files in CSV, TXT, or Excel format
- model.lp
- gurobi.log
- a results output file showing the solution
- readme.txt
- the complete raw LaTeX project files

The Gurobi log must be submitted as a separate file. Its interpretation should also be discussed briefly in the mini paper.

The readme.txt file must clearly state:

- how to run the model
- what each submitted file contains
- the Python version used
- the Gurobi version used
- any additional instructions needed to reproduce the results

The submission deadline is May 20, 2026, at 23:59.